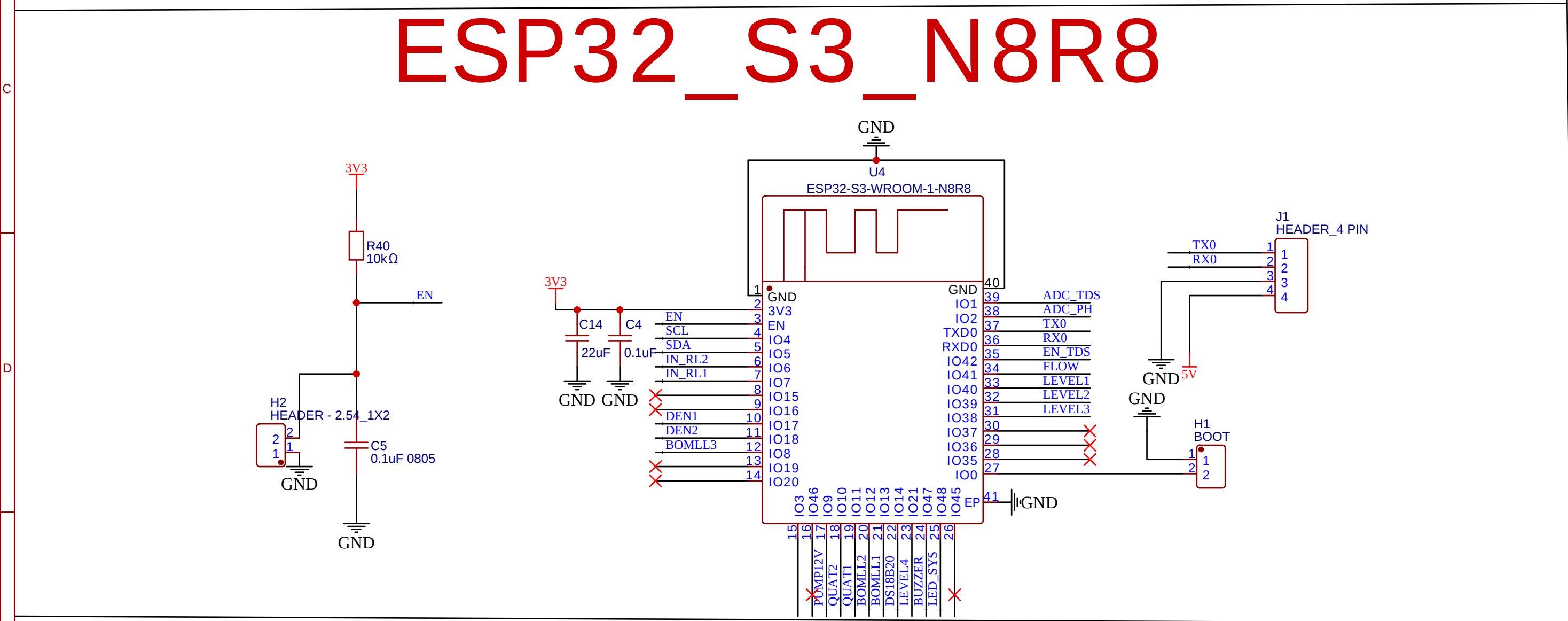
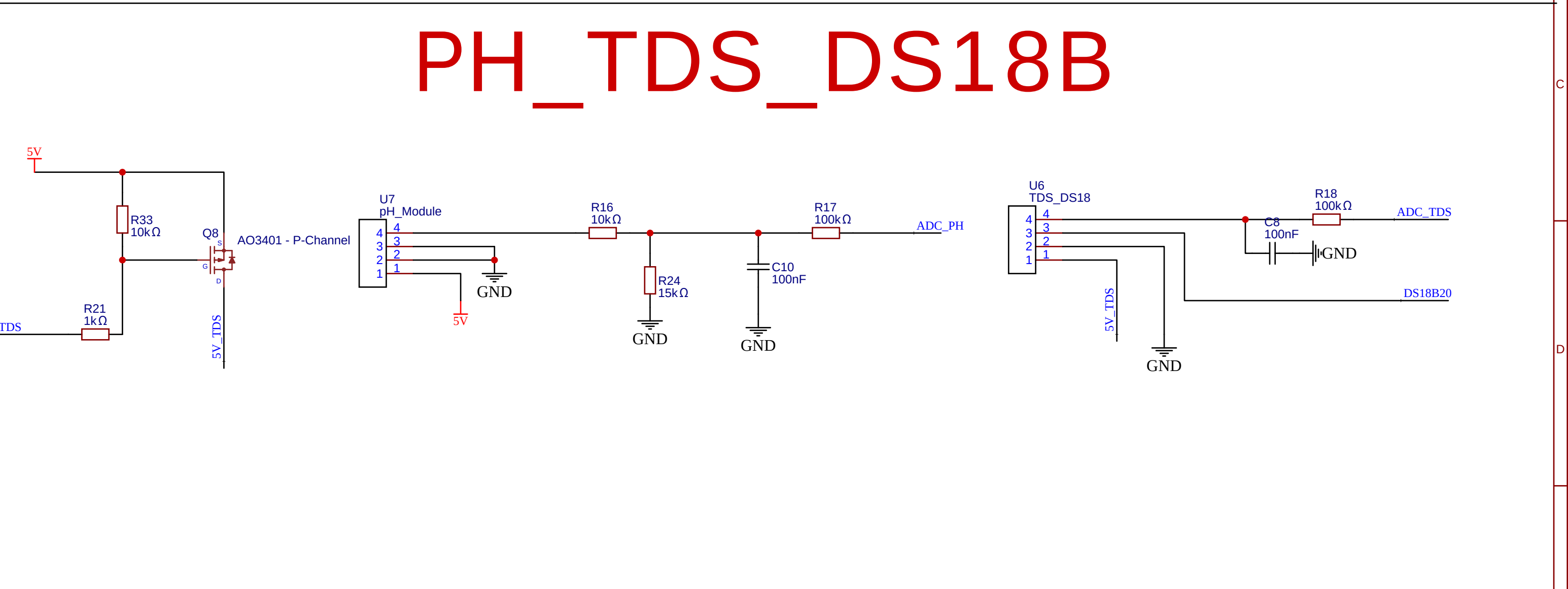


KHỐI NGUỒN



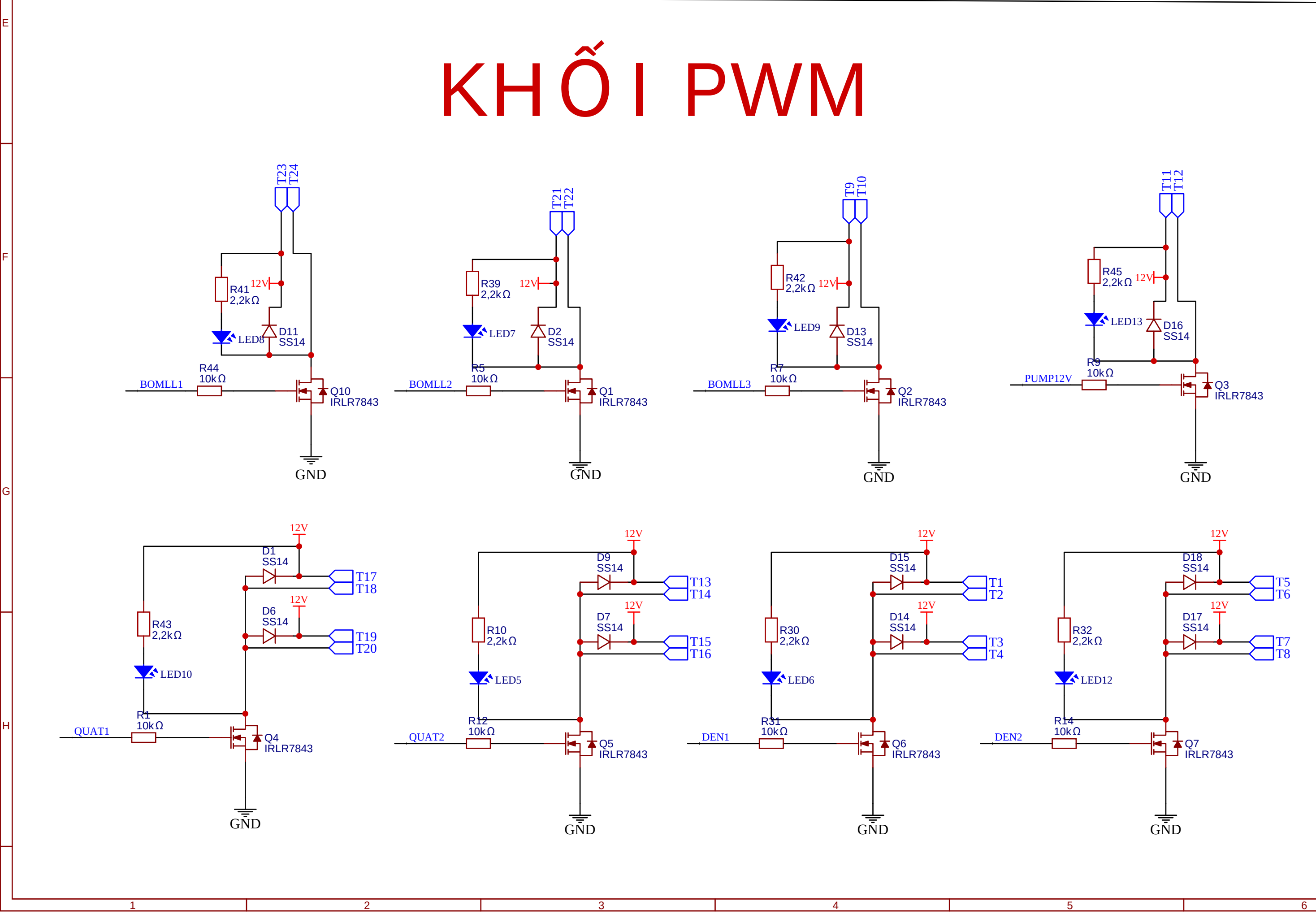
KHOẢNG RELAY

The diagram illustrates a relay control system with two channels. Each channel includes an input stage (IN RL1, IN RL2) and an output stage (RL1, RL2). The input stage uses a 1kΩ resistor (R37, R48) and a 2SC1815G transistor (Q2). The output stage also uses a 2SC1815G transistor (Q2) and a 10kΩ resistor (R46, R47). The relay coil is connected to a 12V source (CN8, CN4) and a 220Ω resistor (R35, R36). The relay switch is connected to a 220V AC source (CN8, CN4).



KHỐI PWM

The image displays eight individual PWM control circuit diagrams, arranged in two rows of four. Each circuit is designed to drive a specific actuator through a MOSFET (Q1-Q8) and a diode (D1-D8). The actuators are represented by blue symbols: T23/T24, T21/T22, T9/T10, T11/T12, T17/T18, T13/T14, T19/T20, T1/T2, T3/T4, T5/T6, T7/T8, and T11/T12. The circuits are labeled with their respective input signals: BOMLL1, BOMLL2, BOMLL3, PUMP12V, QUAT1, QUAT2, DEN1, and DEN2. Each circuit includes a 12V supply, a 2.2kΩ resistor (R41-R48), a 10kΩ resistor (R44-R48), a MOSFET (Q1-Q8, IRLR7843), and a diode (D1-D8, SS14). The output of each circuit is connected to a specific actuator. The top row shows four circuits, and the bottom row shows four circuits, all connected to GND.



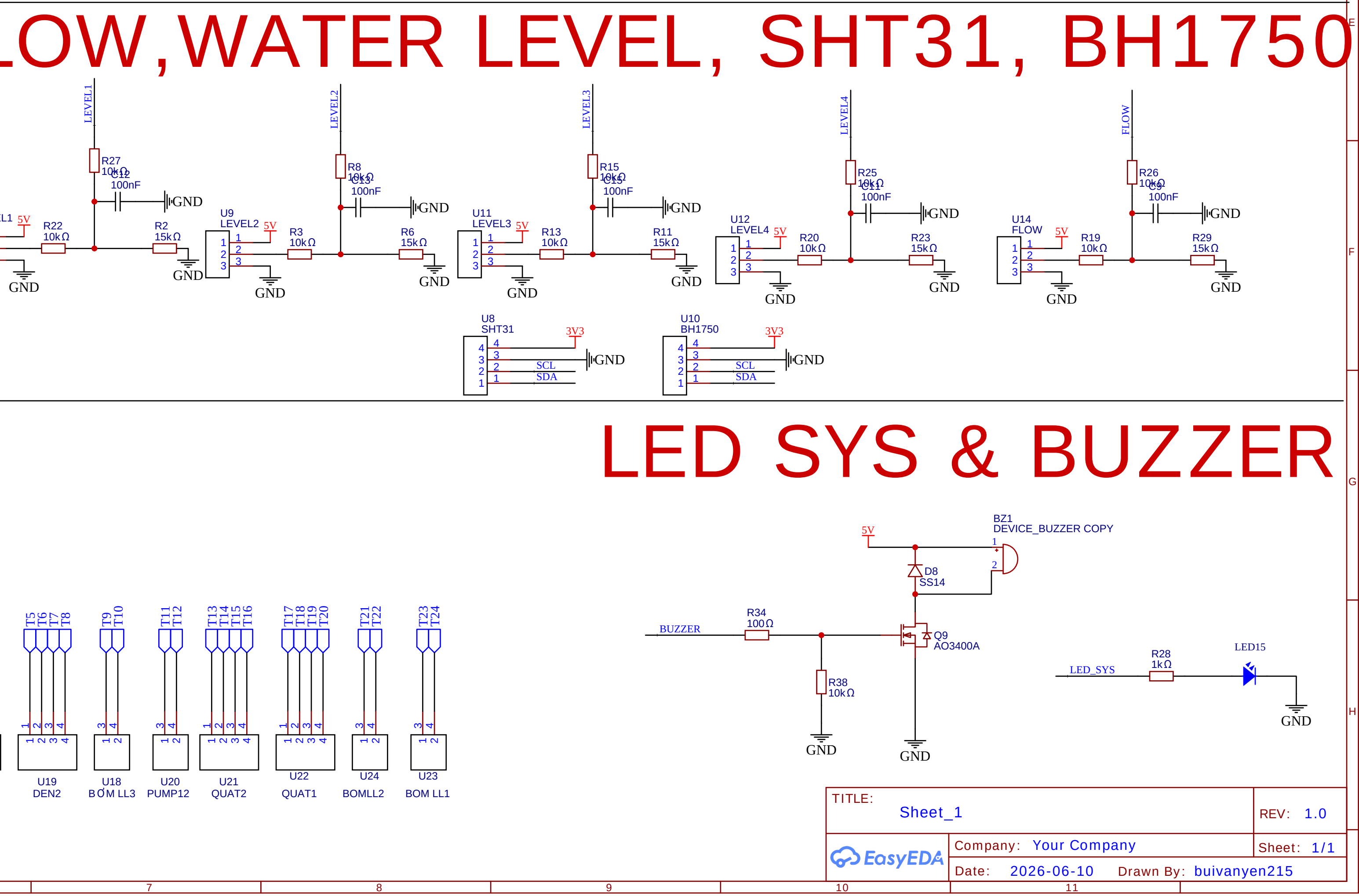
LOW, WATER LEVEL, SHT31, BH1750

The circuit diagram illustrates the hardware setup for monitoring low water levels and environmental conditions. Five input channels (LEVEL1 to FLOW) are connected to a microcontroller via pull-up resistors (R22-R26), decoupling capacitors (10kΩ, 100nF), and logic inverters (U9-U14). The microcontroller also interfaces with an SHT31 sensor (U8) and a BH1750 sensor (U10) via I2C/SMBus. Pin headers U19 through U23 are shown at the bottom left.

LED SYS & BUZZER

The circuit diagram illustrates the hardware setup for the LED system and buzzer. A buzzer (BUZZER) is connected to a microcontroller pin through a 100Ω resistor (R34). A 5V supply is connected to the buzzer's other terminal through a 10kΩ resistor (R38). An LED (LED15) is connected to the buzzer's output through a 1kΩ resistor (R28).

TITLE:		REV: 1.0
Sheet_1		Sheet: 1/1
Company: Your Company		Date: 2026-06-10 Drawn By: buivanyen215



LED SYS & BUZZER

The diagram illustrates the electrical connections for an LED system and a buzzer. It features a 5V power supply connected to a buzzer (BZ1) through a 100Ω resistor (R34). A 10kΩ resistor (R38) is connected to ground. An LED (LED15) is connected to a 1kΩ resistor (R28) and ground. The LED is connected to a 5V supply. The diagram also shows a 5V supply connected to a 100Ω resistor (R34) and a 10kΩ resistor (R38) connected to ground. A 5V supply is also connected to a 1kΩ resistor (R28) and ground. The diagram includes a title block with the following information: TITLE: Sheet_1, REV: 1.0, Company: Your Company, Sheet: 1/1, Date: 2026-06-10, Drawn By: buivanyen215.

